Online Retail Analysis

Pretty

2026-04-05

```r
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)
```

Business Problem

The objective of this analysis is to understand customer purchasing behaviour in an online retail environment. Specifically, the analysis aims to answer:

- Who are the key customers?
- What products do they purchase?
- How does purchasing behaviour differ across countries and product types?

Load Required Libraries and Dataset

```r
library(data.table)
library(ggplot2)
library(readr)

filePath <- "C:/Users/Owner/Desktop/Portfolio/Product Analysis-R/online+retail/"
data <- fread(paste0(filePath,"OnlineRetail.csv"))
```

Exploratory Data Analysis

The first step in any analysis is to understand the structure and quality of the data.

```r
# Examine the data
str(data)
```

```
## Classes 'data.table' and 'data.frame':   541909 obs. of  8 variables:
##  $ InvoiceNo  : chr  "536365" "536365" "536365" "536365" ...
##  $ StockCode  : chr  "85123A" "71053" "84406B" "84029G" ...
##  $ Description: chr  "WHITE HANGING HEART T-LIGHT HOLDER" "WHITE METAL LANTERN" "CREAM CUPID HEARTS C
##  $ Quantity   : int  6 6 8 6 6 2 6 6 6 32 ...
##  $ InvoiceDate: chr  "01/12/2010 08:26" "01/12/2010 08:26" "01/12/2010 08:26" "01/12/2010 08:26" ...
##  $ UnitPrice  : num  2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85 1.85 1.69 ...
##  $ CustomerID : int  17850 17850 17850 17850 17850 17850 17850 17850 17850 13047 ...
##  $ Country    : chr  "United Kingdom" "United Kingdom" "United Kingdom" "United Kingdom" ...
##  - attr(*, ".internal.selfref")=<externalptr>
```

Insight: The dataset contains over 500,000 transactions with key variables such as product description, quantity, price, customer ID, and country.

```r
# Summarise the data to check for possible null and outliers
summary(data)
```

```
##   InvoiceNo          StockCode         Description           Quantity
##  Length:541909      Length:541909      Length:541909      Min.   :-80995.000
##  Class :character   Class :character   Class :character   1st Qu.:     1.000
##  Mode  :character   Mode  :character   Mode  :character   Median :     3.000
##                                                           Mean   :     9.552
##                                                           3rd Qu.:    10.000
##                                                           Max.   : 80995.000
##
##  InvoiceDate          UnitPrice            CustomerID       Country
##  Length:541909      Min.   :-11062.060   Min.   :12346    Length:541909
##  Class :character   1st Qu.:     1.250   1st Qu.:13953    Class :character
##  Mode  :character   Median :     2.080   Median :15152    Mode  :character
##                     Mean   :     4.611   Mean   :15288
##                     3rd Qu.:     4.130   3rd Qu.:16791
##                     Max.   : 38970.000   Max.   :18287
##                                          NA's   :135080
```

Key Observations:

- Negative values in Quantity and UnitPrice indicate returns or data errors
- Missing values in CustomerID (~135,000 records)
- Large variation in quantities suggests potential outliers

Clean the Data

To ensure accurate analysis, the dataset was cleaned as follows:

```r
# Remove missing customers
data <- data[!is.na(CustomerID)]

# Remove negative or zero quantities
data <- data[Quantity > 0]

# Remove negative or zero prices
data <- data[UnitPrice > 0]
```

Insight: Removing invalid transactions ensures that only genuine purchases are analysed.

Feature Engineering

New variables are creted to support analysis

Convert the Date

```r
data[, InvoiceDate := as.POSIXct(InvoiceDate, format = "%d/%m/%Y %H:%M")]
```

Create Total Sales

```r
data[, TOTAL_SALES := Quantity * UnitPrice]
```

Insight: TOTAL_SALES is a key metric used to measure customer value and revenue contribution.

Product Analysis

```r
data[, .N, by = Description][order(-N)]
```

```
##                              Description     N
##                                   <char> <int>
##    1: WHITE HANGING HEART T-LIGHT HOLDER  2028
##    2:           REGENCY CAKESTAND 3 TIER  1723
##    3:            JUMBO BAG RED RETROSPOT  1618
##    4:      ASSORTED COLOUR BIRD ORNAMENT  1408
##    5:                      PARTY BUNTING  1396
##   ---
## 3862:        LETTER ""T"" BLING KEY RING     1
## 3863:        LETTER ""U"" BLING KEY RING     1
## 3864:        LETTER ""W"" BLING KEY RING     1
## 3865:        LETTER ""Z"" BLING KEY RING     1
## 3866:        PAPER CRAFT , LITTLE BIRDIE     1
```

Insight: Popular products include decorative and gift-related items such as: - WHITE HANGING HEART T-LIGHT HOLDER - REGENCY CAKESTAND The product mix suggests a strong focus on home décor and seasonal items Best-selling products are primarily decorative and gift-oriented.

Text Analysis of Products

```r
productWords <- data.table(unlist(strsplit(unique(data$Description), " ")))
setnames(productWords, "words")

# Remove numbers
productWords <- productWords[!grepl("\\d", words)]

# Keep only alphabetic
productWords <- productWords[grepl("^[A-Za-z]+$", words)]

# Count frequency
productWords[, .N, by = words][order(-N)]
```

```
##           words     N
##          <char> <int>
##    1:      PINK   284
##    2:       SET   278
##    3:        OF   237
##    4:     HEART   234
##    5:   VINTAGE   215
##   ---
## 1945:     CROSS     1
## 1946:  PUNCTURE     1
## 1947:   COUNTRY     1
## 1948: PERFECTLY     1
## 1949:    BIRDIE     1
```

Insight: Frequent words include: PINK, SET, HEART, VINTAGE This indicates strong themes around: - Gifts - Decorations - Aesthetic/home styling

Sales by Country

```r
data[, .(SALES = sum(TOTAL_SALES)), by = Country][order(-SALES)]
```

```
##                  Country      SALES
##                   <char>      <num>
##  1:       United Kingdom 7308391.55
##  2:          Netherlands  285446.34
##  3:                 EIRE  265545.90
##  4:              Germany  228867.14
##  5:               France  209024.05
##  6:            Australia  138521.31
##  7:                Spain   61577.11
##  8:          Switzerland   56443.95
##  9:              Belgium   41196.34
## 10:               Sweden   38378.33
## 11:                Japan   37416.37
## 12:               Norway   36165.44
## 13:             Portugal   33439.89
## 14:              Finland   22546.08
## 15:            Singapore   21279.29
## 16:      Channel Islands   20450.44
## 17:              Denmark   18955.34
## 18:                Italy   17483.24
## 19:               Cyprus   13590.38
## 20:              Austria   10198.68
## 21:               Poland    7334.65
## 22:               Israel    7221.69
## 23:               Greece    4760.52
## 24:              Iceland    4310.00
## 25:               Canada    3666.38
## 26:                  USA    3580.39
## 27:                Malta    2725.59
## 28:          Unspecified    2667.07
```

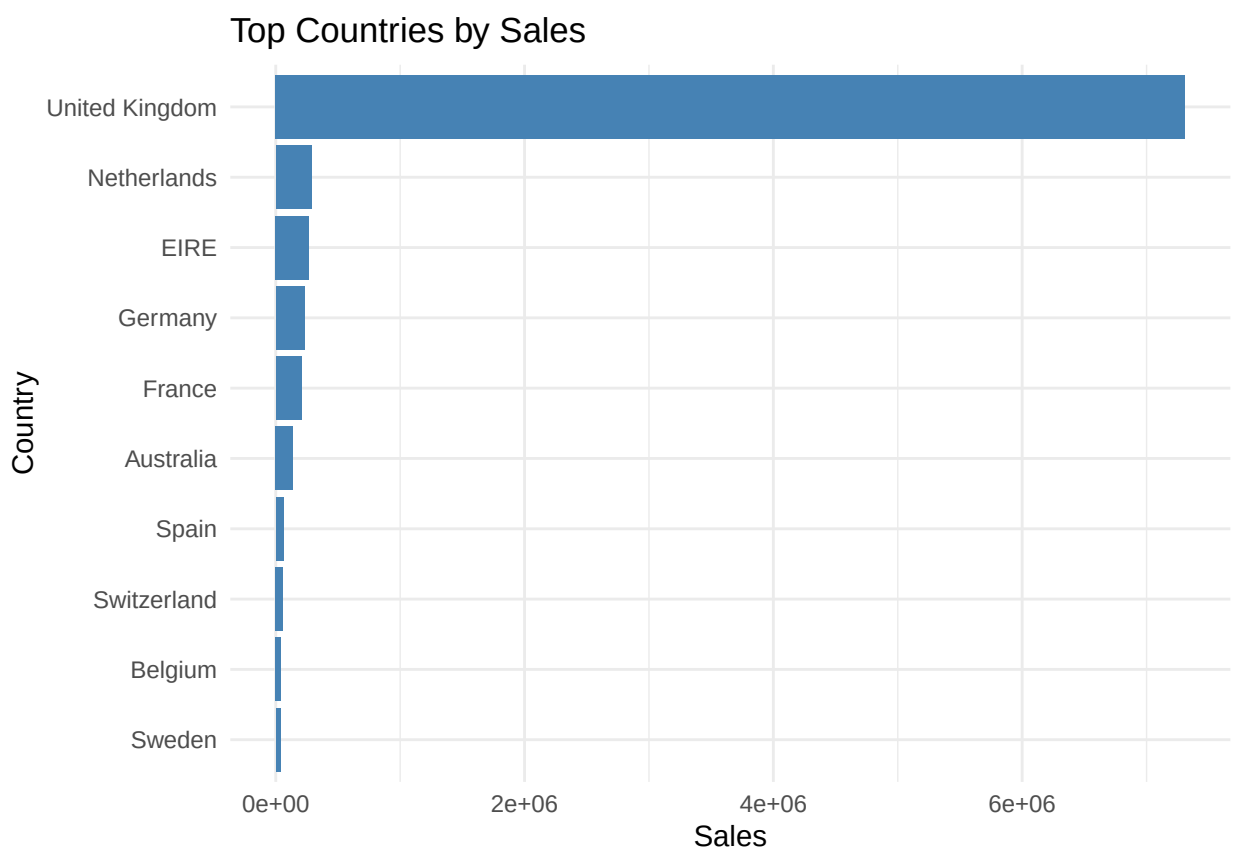
```
## 29: United Arab Emirates    1902.28
## 30:              Lebanon    1693.88
## 31:            Lithuania    1661.06
## 32:   European Community    1300.25
## 33:               Brazil    1143.60
## 34:                  RSA    1002.31
## 35:       Czech Republic     826.74
## 36:              Bahrain     548.40
## 37:         Saudi Arabia     145.92
##                  Country      SALES
##                   <char>      <num>
```

Key Insights: - The United Kingdom dominates sales significantly Secondary markets include: - Netherlands - Ireland (EIRE) - Germany and France International markets contribute smaller but meaningful revenue streams

```
top_countries <- data[, .(SALES = sum(TOTAL_SALES)), by = Country][order(-SALES)][1:10]

p <- ggplot(top_countries,
aes(x = reorder(Country, SALES), y = SALES)) +
geom_bar(stat = "identity", fill = "steelblue") +
coord_flip() +
labs(title = "Top Countries by Sales", x = "Country", y = "Sales") +
theme_minimal()

print(p)
```



Insight: The United Kingdom dominates sales, followed by Netherlands, EIRE, Germany, and France.

Customer Analysis

```
# Number of Customers by Country
data[, .(CUSTOMERS = uniqueN(CustomerID)), by = Country][order(-CUSTOMERS)]
```

```
##                  Country CUSTOMERS
##                   <char>     <int>
##  1:       United Kingdom      3920
##  2:              Germany        94
##  3:               France        87
##  4:                Spain        30
##  5:              Belgium        25
##  6:          Switzerland        21
##  7:             Portugal        19
##  8:                Italy        14
##  9:              Finland        12
## 10:              Austria        11
## 11:               Norway        10
## 12:            Australia         9
## 13:          Netherlands         9
## 14:      Channel Islands         9
## 15:              Denmark         9
## 16:                Japan         8
## 17:               Cyprus         8
## 18:               Sweden         8
## 19:               Poland         6
## 20:               Greece         4
## 21:               Canada         4
## 22:          Unspecified         4
## 23:                  USA         4
## 24:                 EIRE         3
## 25:               Israel         3
## 26: United Arab Emirates         2
## 27:              Bahrain         2
## 28:                Malta         2
## 29:            Lithuania         1
## 30:              Iceland         1
## 31:            Singapore         1
## 32:              Lebanon         1
## 33:         Saudi Arabia         1
## 34:       Czech Republic         1
## 35:               Brazil         1
## 36:   European Community         1
## 37:                  RSA         1
##                  Country CUSTOMERS
##                   <char>     <int>
```

Insight: The UK has the largest customer base Other countries have relatively small but diverse customer groups

Key Customer Identification

```
top_customers <- data[, .(TOTAL_SPEND = sum(TOTAL_SALES)), by = CustomerID][order(-TOTAL_SPEND)][1:10]

top_customers
```

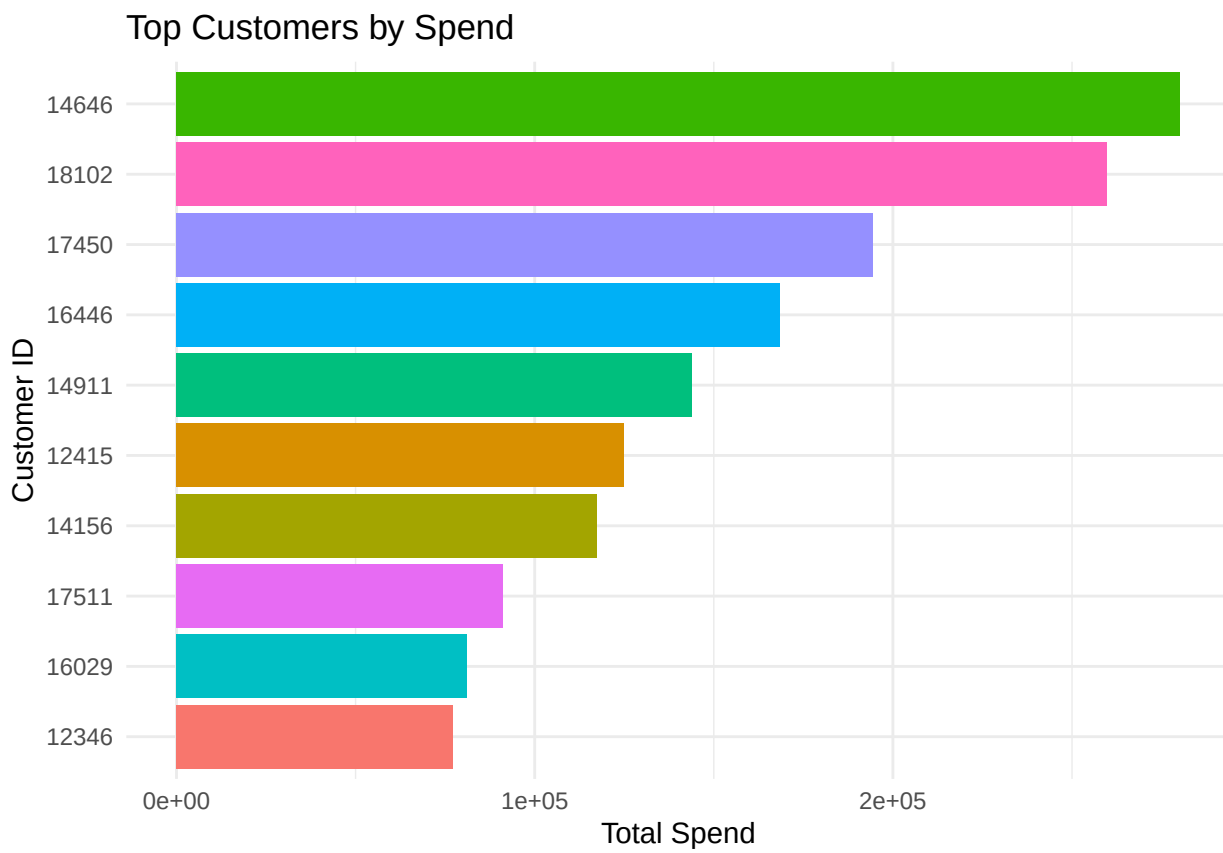
```
##     CustomerID TOTAL_SPEND
##          <int>       <num>
##  1:      14646   280206.02
##  2:      18102   259657.30
##  3:      17450   194550.79
##  4:      16446   168472.50
##  5:      14911   143825.06
##  6:      12415   124914.53
##  7:      14156   117379.63
##  8:      17511    91062.38
##  9:      16029    81024.84
## 10:      12346    77183.60
```

Insight: A small number of customers contribute a large share of revenue (Pareto principle).

Visuals for Top Customers

```
p_top_customers <- ggplot(
  top_customers,
  aes(x = reorder(as.factor(CustomerID), TOTAL_SPEND),
      y = TOTAL_SPEND,
      fill = as.factor(CustomerID))
) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(
    title = "Top Customers by Spend",
    x = "Customer ID",
    y = "Total Spend"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

print(p_top_customers)
```




```
ggsave("plots/top_customers_coloured.png", plot = p_top_customers, width = 8, height = 6)
```

```
# Average Spend per Customer
data[, .(AVG_SPEND = sum(TOTAL_SALES)/uniqueN(CustomerID)), by = Country]
```

```
##                   Country   AVG_SPEND
##                    <char>       <num>
##  1:        United Kingdom   1864.3856
##  2:                France   2402.5753
##  3:             Australia  15391.2567
##  4:           Netherlands  31716.2600
##  5:               Germany   2434.7568
##  6:                Norway   3616.5440
##  7:                  EIRE  88515.3000
##  8:           Switzerland   2687.8071
##  9:                 Spain   2052.5703
## 10:                Poland   1222.4417
## 11:              Portugal   1759.9942
## 12:                 Italy   1248.8029
## 13:               Belgium   1647.8536
## 14:             Lithuania   1661.0600
## 15:                 Japan   4677.0463
## 16:               Iceland   4310.0000
## 17:       Channel Islands   2272.2711
## 18:               Denmark   2106.1489
```

```
## 19:               Cyprus  1698.7975
## 20:               Sweden  4797.2913
## 21:              Finland  1878.8400
## 22:              Austria   927.1527
## 23:               Greece  1190.1300
## 24:            Singapore 21279.2900
## 25:              Lebanon  1693.8800
## 26: United Arab Emirates   951.1400
## 27:               Israel  2407.2300
## 28:         Saudi Arabia   145.9200
## 29:       Czech Republic   826.7400
## 30:               Canada   916.5950
## 31:          Unspecified   666.7675
## 32:               Brazil  1143.6000
## 33:                  USA   895.0975
## 34:   European Community  1300.2500
## 35:              Bahrain   274.2000
## 36:                Malta  1362.7950
## 37:                  RSA  1002.3100
##                  Country  AVG_SPEND
##                   <char>      <num>
```

Insight: Some countries (e.g. Netherlands, EIRE) show very high average spend This suggests: - Bulk purchasing behaviour - Possibly business or wholesale customer

Purchasing Behaviour

```r
# Units per customer
data[, .(AVG_UNITS = sum(Quantity)/uniqueN(CustomerID)), by = Country]
```

```
##                  Country  AVG_UNITS
##                   <char>      <num>
##  1:       United Kingdom  1085.9031
##  2:               France  1281.2759
##  3:            Australia  9322.3333
##  4:          Netherlands 22262.3333
##  5:              Germany  1268.7340
##  6:               Norway  1933.6000
##  7:                 EIRE 46758.3333
##  8:          Switzerland  1432.4762
##  9:                Spain   931.3333
## 10:               Poland   614.0000
## 11:             Portugal   848.5263
## 12:                Italy   579.4286
## 13:              Belgium   929.4800
## 14:            Lithuania   652.0000
## 15:                Japan  3252.0000
## 16:              Iceland  2458.0000
## 17:      Channel Islands  1054.5556
## 18:              Denmark   915.0000
## 19:               Cyprus   795.1250
```

```
## 20:                Sweden  4510.3750
## 21:               Finland   892.0000
## 22:               Austria   443.7273
## 23:                Greece   389.2500
## 24:             Singapore  5241.0000
## 25:               Lebanon   386.0000
## 26: United Arab Emirates   491.0000
## 27:                Israel  1348.6667
## 28:          Saudi Arabia    80.0000
## 29:        Czech Republic   671.0000
## 30:                Canada   690.7500
## 31:           Unspecified   447.2500
## 32:                Brazil   356.0000
## 33:                   USA   614.5000
## 34:   European Community   499.0000
## 35:               Bahrain   130.0000
## 36:                 Malta   485.0000
## 37:                   RSA   351.0000
##                   Country  AVG_UNITS
##                    <char>      <num>
```

Insight: - Countries like Netherlands and EIRE purchase significantly higher volumes - Indicates high-volume buyers or commercial activity

```r
# Average Price per Unit
data[, .(AVG_PRICE = sum(TOTAL_SALES)/sum(Quantity)), by = Country]
```

```
##                   Country AVG_PRICE
##                    <char>     <num>
##  1:       United Kingdom  1.716899
##  2:               France  1.875143
##  3:            Australia  1.651009
##  4:          Netherlands  1.424660
##  5:              Germany  1.919044
##  6:               Norway  1.870368
##  7:                 EIRE  1.893038
##  8:          Switzerland  1.876336
##  9:                Spain  2.203905
## 10:               Poland  1.990947
## 11:             Portugal  2.074178
## 12:                Italy  2.155232
## 13:              Belgium  1.772877
## 14:            Lithuania  2.547638
## 15:                Japan  1.438206
## 16:              Iceland  1.753458
## 17:      Channel Islands  2.154719
## 18:              Denmark  2.301802
## 19:               Cyprus  2.136516
## 20:               Sweden  1.063613
## 21:              Finland  2.106323
## 22:              Austria  2.089465
## 23:               Greece  3.057495
## 24:            Singapore  4.060158
```

```
## 25:              Lebanon  4.388290
## 26: United Arab Emirates  1.937149
## 27:               Israel  1.784896
## 28:         Saudi Arabia  1.824000
## 29:       Czech Republic  1.232101
## 30:               Canada  1.326956
## 31:          Unspecified  1.490816
## 32:               Brazil  3.212360
## 33:                  USA  1.456627
## 34:   European Community  2.605711
## 35:              Bahrain  2.109231
## 36:                Malta  2.809887
## 37:                  RSA  2.855584
##                  Country AVG_PRICE
##                   <char>     <num>
```

Insight: - Price sensitivity varies by country - Some markets (e.g. Singapore, Lebanon) show higher willingness to pay

Dipe Deep: UK vs Other Countries

```r
# Pick the UK
uk <- data[Country == "United Kingdom"]
other <- data[Country != "United Kingdom"]
```

```r
# Product Affinity Analysis
qty_uk <- uk[, sum(Quantity)]
qty_other <- other[, sum(Quantity)]

uk_products <- uk[, .(uk_share = sum(Quantity)/qty_uk), by = Description]
other_products <- other[, .(other_share = sum(Quantity)/qty_other), by = Description]

product_affinity <- merge(uk_products, other_products)
product_affinity[, affinity := uk_share / other_share]

product_affinity[order(-affinity)]
```

```
##                           Description     uk_share  other_share     affinity
##                                <char>        <num>        <num>        <num>
##    1:       VINTAGE BILLBOARD TEA MUG 4.385986e-04 1.097608e-06 3.995949e+02
##    2:   DECORATIVE WICKER HEART SMALL 3.500331e-04 2.195216e-06 1.594527e+02
##    3: KINGS CHOICE GIANT TUBE MATCHES 1.714927e-04 1.097608e-06 1.562422e+02
##    4: LARGE ZINC HEART WALL ORGANISER 2.494867e-04 2.195216e-06 1.136502e+02
##    5:     MIXED NUTS LIGHT GREEN BOWL 1.099433e-04 1.097608e-06 1.001663e+02
##   ---
## 2847:    BLUE BUNNY EASTER EGG BASKET 3.053980e-06 1.580556e-04 1.932219e-02
## 2848:          TURQ ICE CREAM BUM BAG 2.349216e-07 1.317130e-05 1.783587e-02
## 2849:     DOORKNOB CRACKED GLAZE BLUE 2.819059e-06 1.646412e-04 1.712244e-02
## 2850:                         POSTAGE 8.927019e-06 3.382828e-03 2.638922e-03
## 2851:                     RAIN PONCHO 9.396862e-07 4.467265e-04 2.103493e-03
```

Key Insights: UK customers show strong preference for: - Vintage and decorative items Some products are significantly more popular in the UK compared to other regions This reflects regional taste differences and localisation opportunities

Deep Dive: Identifying Growth Opportunities (Excluding UK)

The United Kingdom dominates the dataset, which can mask insights from smaller but potentially high-value markets. To uncover international growth opportunities, the UK is excluded from this analysis.

Create Non-UK Dataset

```
non_uk <- data[Country != "United Kingdom"]
```

Sales by Country (Non-UK)

```
non_uk[, .(SALES = sum(TOTAL_SALES)), by = Country][order(-SALES)]
```

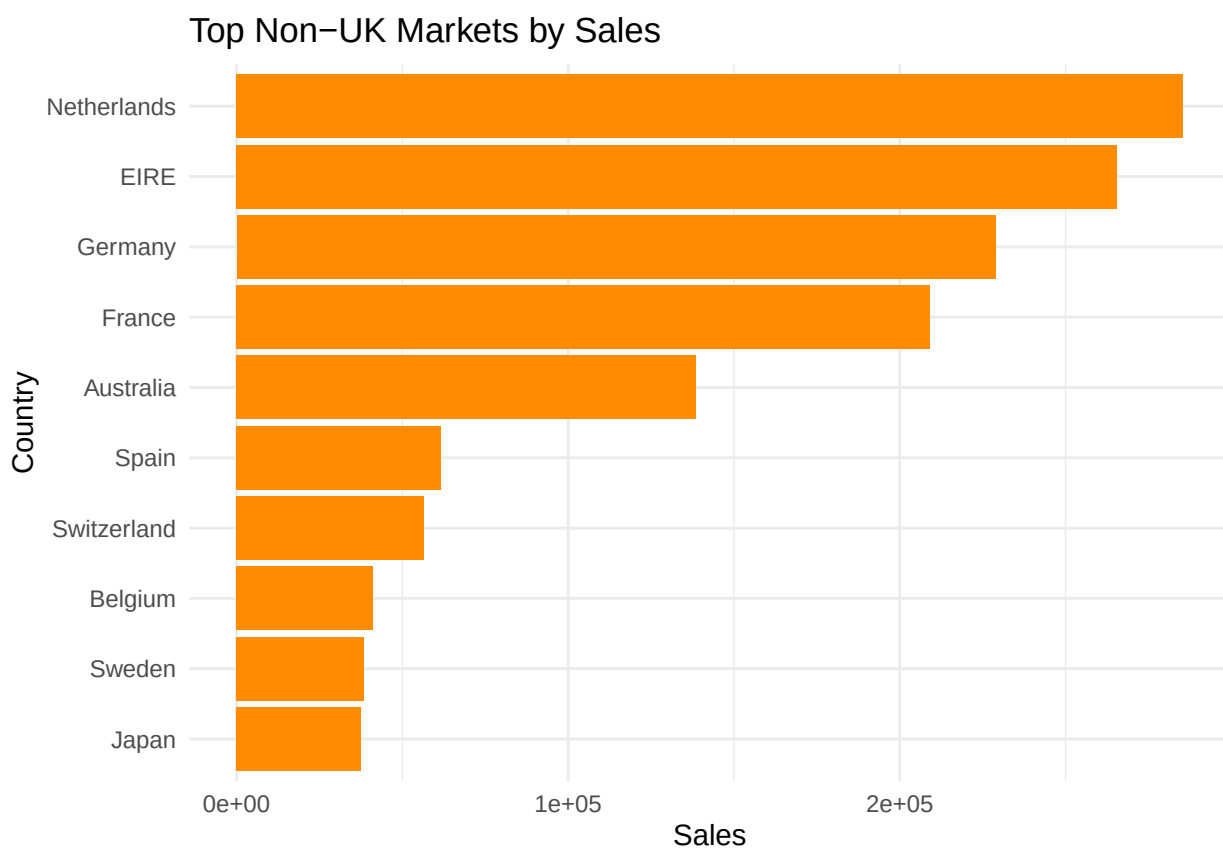
```
##                  Country     SALES
##                   <char>     <num>
##  1:          Netherlands 285446.34
##  2:                 EIRE 265545.90
##  3:              Germany 228867.14
##  4:               France 209024.05
##  5:            Australia 138521.31
##  6:                Spain  61577.11
##  7:          Switzerland  56443.95
##  8:              Belgium  41196.34
##  9:               Sweden  38378.33
## 10:                Japan  37416.37
## 11:               Norway  36165.44
## 12:             Portugal  33439.89
## 13:              Finland  22546.08
## 14:            Singapore  21279.29
## 15:      Channel Islands  20450.44
## 16:              Denmark  18955.34
## 17:                Italy  17483.24
## 18:               Cyprus  13590.38
## 19:              Austria  10198.68
## 20:               Poland   7334.65
## 21:               Israel   7221.69
## 22:               Greece   4760.52
## 23:              Iceland   4310.00
## 24:               Canada   3666.38
## 25:                  USA   3580.39
## 26:                Malta   2725.59
## 27:          Unspecified   2667.07
## 28: United Arab Emirates   1902.28
## 29:              Lebanon   1693.88
```

```
## 30:            Lithuania   1661.06
## 31:   European Community   1300.25
## 32:               Brazil   1143.60
## 33:                  RSA   1002.31
## 34:       Czech Republic    826.74
## 35:              Bahrain    548.40
## 36:         Saudi Arabia    145.92
##                  Country     SALES
##                   <char>     <num>
```

```
non_uk_sales <- non_uk[, .(SALES = sum(TOTAL_SALES)), by = Country][order(-SALES)][1:10]

p <- ggplot(non_uk_sales, aes(x = reorder(Country, SALES), y = SALES)) +
  geom_bar(stat = "identity", fill = "darkorange") +
  coord_flip() +
  labs(title = "Top Non-UK Markets by Sales", x = "Country", y = "Sales") +
  theme_minimal()

print(p)
```




```
ggsave("plots/top_non_uk_markets.png", plot = p, width = 8, height = 6)
```

Insight: Netherlands, EIRE, and Germany are the strongest international markets.

High Value Customer

```
non_uk[, .(AVG_SPEND = sum(TOTAL_SALES)/uniqueN(CustomerID)),
       by = Country][order(-AVG_SPEND)]
```

```
##                  Country  AVG_SPEND
##                   <char>      <num>
##  1:                 EIRE 88515.3000
##  2:          Netherlands 31716.2600
##  3:            Singapore 21279.2900
##  4:            Australia 15391.2567
##  5:               Sweden  4797.2913
##  6:                Japan  4677.0463
##  7:              Iceland  4310.0000
##  8:               Norway  3616.5440
##  9:          Switzerland  2687.8071
## 10:              Germany  2434.7568
## 11:               Israel  2407.2300
## 12:               France  2402.5753
## 13:      Channel Islands  2272.2711
## 14:              Denmark  2106.1489
## 15:                Spain  2052.5703
## 16:              Finland  1878.8400
## 17:             Portugal  1759.9942
## 18:               Cyprus  1698.7975
## 19:              Lebanon  1693.8800
## 20:            Lithuania  1661.0600
## 21:              Belgium  1647.8536
## 22:                Malta  1362.7950
## 23:   European Community  1300.2500
## 24:                Italy  1248.8029
## 25:               Poland  1222.4417
## 26:               Greece  1190.1300
## 27:               Brazil  1143.6000
## 28:                  RSA  1002.3100
## 29: United Arab Emirates   951.1400
## 30:              Austria   927.1527
## 31:               Canada   916.5950
## 32:                  USA   895.0975
## 33:       Czech Republic   826.7400
## 34:          Unspecified   666.7675
## 35:              Bahrain   274.2000
## 36:         Saudi Arabia   145.9200
##                  Country  AVG_SPEND
##                   <char>      <num>
```

Volume Behaviour

```
non_uk[, .(AVG_UNITS = sum(Quantity)/uniqueN(CustomerID)),
       by = Country][order(-AVG_UNITS)]
```

```
##                   Country  AVG_UNITS
##                    <char>      <num>
##  1:                  EIRE 46758.3333
##  2:           Netherlands 22262.3333
##  3:             Australia  9322.3333
##  4:             Singapore  5241.0000
##  5:                Sweden  4510.3750
##  6:                 Japan  3252.0000
##  7:               Iceland  2458.0000
##  8:                Norway  1933.6000
##  9:           Switzerland  1432.4762
## 10:                Israel  1348.6667
## 11:                France  1281.2759
## 12:               Germany  1268.7340
## 13:       Channel Islands  1054.5556
## 14:                 Spain   931.3333
## 15:               Belgium   929.4800
## 16:               Denmark   915.0000
## 17:               Finland   892.0000
## 18:              Portugal   848.5263
## 19:                Cyprus   795.1250
## 20:                Canada   690.7500
## 21:        Czech Republic   671.0000
## 22:             Lithuania   652.0000
## 23:                   USA   614.5000
## 24:                Poland   614.0000
## 25:                 Italy   579.4286
## 26:    European Community   499.0000
## 27: United Arab Emirates   491.0000
## 28:                 Malta   485.0000
## 29:           Unspecified   447.2500
## 30:               Austria   443.7273
## 31:                Greece   389.2500
## 32:               Lebanon   386.0000
## 33:                Brazil   356.0000
## 34:                   RSA   351.0000
## 35:               Bahrain   130.0000
## 36:          Saudi Arabia    80.0000
##                   Country  AVG_UNITS
##                    <char>      <num>
```

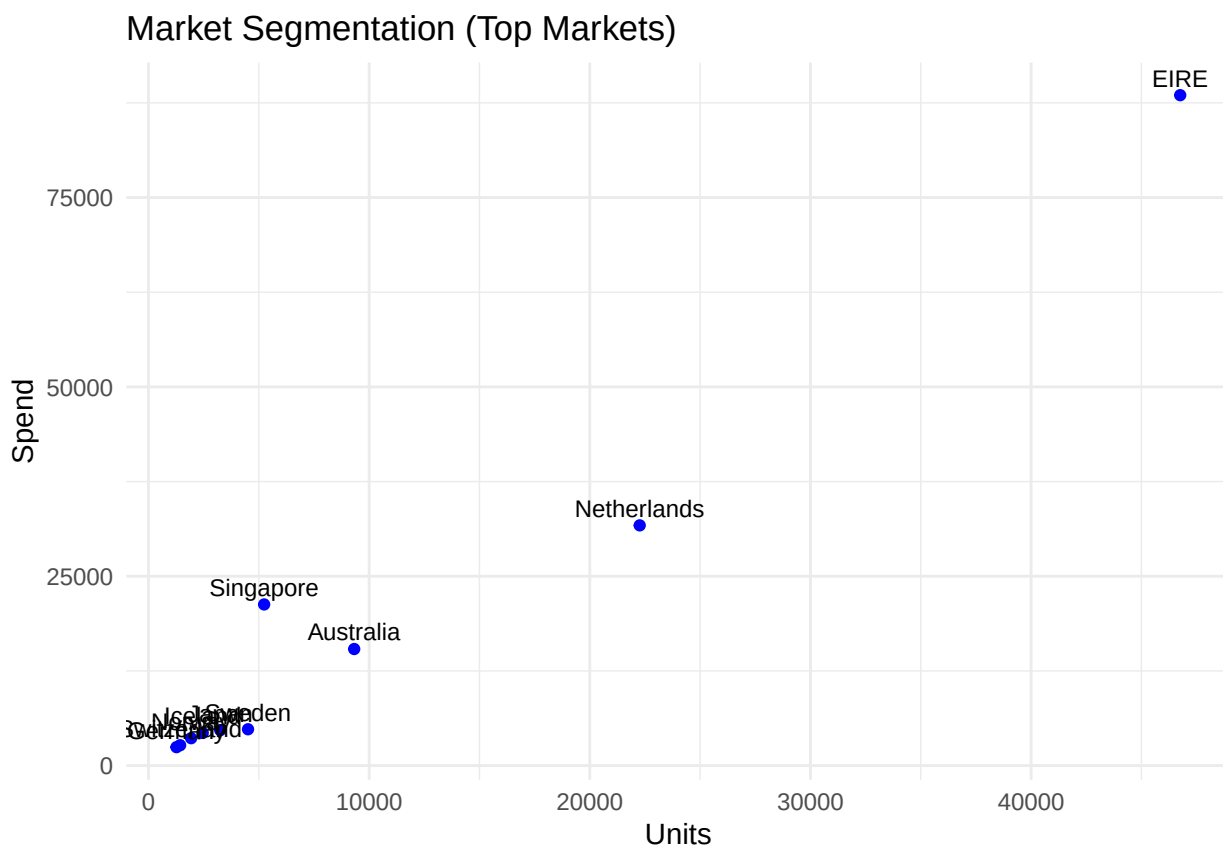
Market Segmentation: Volume vs Value

```r
market_profile <- non_uk[, .(
  AVG_SPEND = sum(TOTAL_SALES)/uniqueN(CustomerID),
  AVG_UNITS = sum(Quantity)/uniqueN(CustomerID)
), by = Country]

top_markets <- market_profile[order(-AVG_SPEND)][1:10]

ggplot(top_markets, aes(x = AVG_UNITS, y = AVG_SPEND)) +
geom_point(color = "blue") +
```

```r
geom_text(aes(label = Country), vjust = -0.5, size = 3) +
labs(title = "Market Segmentation (Top Markets)", x = "Units", y = "Spend") +
theme_minimal()
```



Insight: - High Volume + High Value → Wholesale markets
- Low Volume + High Value → Premium markets
- Low Volume + Low Value → Emerging markets

Price Sensitivity

```r
non_uk[, .(AVG_PRICE = sum(TOTAL_SALES)/sum(Quantity)),
       by = Country][order(-AVG_PRICE)]
```

```
##                  Country AVG_PRICE
##                   <char>     <num>
##  1:               Lebanon  4.388290
##  2:             Singapore  4.060158
##  3:                Brazil  3.212360
##  4:                Greece  3.057495
##  5:                   RSA  2.855584
##  6:                 Malta  2.809887
##  7:   European Community  2.605711
##  8:             Lithuania  2.547638
```

```
##  9:              Denmark  2.301802
## 10:                Spain  2.203905
## 11:                Italy  2.155232
## 12:      Channel Islands  2.154719
## 13:               Cyprus  2.136516
## 14:              Bahrain  2.109231
## 15:              Finland  2.106323
## 16:              Austria  2.089465
## 17:             Portugal  2.074178
## 18:               Poland  1.990947
## 19: United Arab Emirates  1.937149
## 20:              Germany  1.919044
## 21:                 EIRE  1.893038
## 22:          Switzerland  1.876336
## 23:               France  1.875143
## 24:               Norway  1.870368
## 25:         Saudi Arabia  1.824000
## 26:               Israel  1.784896
## 27:              Belgium  1.772877
## 28:              Iceland  1.753458
## 29:            Australia  1.651009
## 30:          Unspecified  1.490816
## 31:                  USA  1.456627
## 32:                Japan  1.438206
## 33:          Netherlands  1.424660
## 34:               Canada  1.326956
## 35:       Czech Republic  1.232101
## 36:               Sweden  1.063613
##                  Country AVG_PRICE
##                   <char>     <num>
```

Product Opportunity

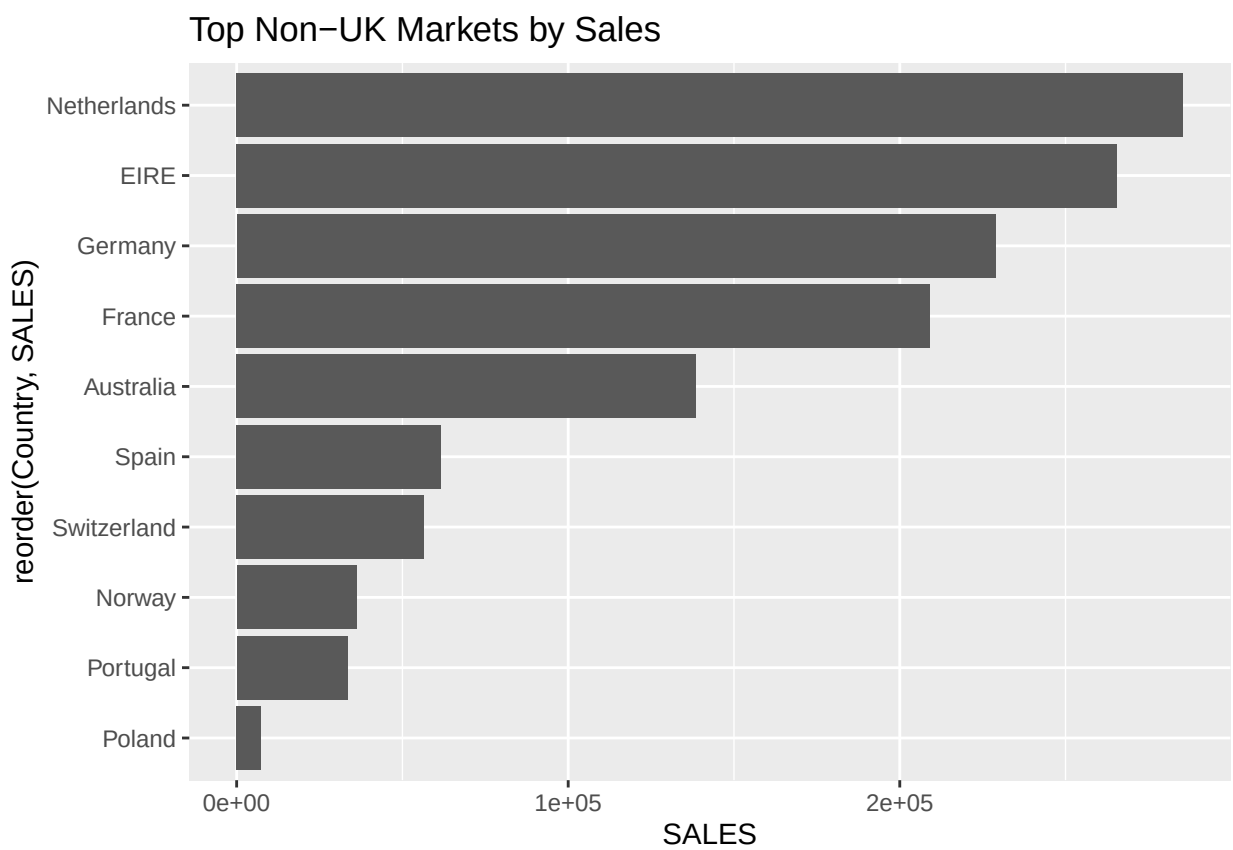
```r
non_uk_products <- non_uk[, .(QTY = sum(Quantity)), by = Description]
non_uk_products[order(-QTY)]
```

```
##                           Description   QTY
##                                <char> <int>
##    1:              RABBIT NIGHT LIGHT 15437
##    2:          MINI PAINT SET VINTAGE 12493
##    3: PACK OF 72 RETROSPOT CAKE CASES 11228
##    4:              SPACEBOY LUNCH BOX  8364
##    5:            DOLLY GIRL LUNCH BOX  7344
##   ---
## 2880:           High Resolution Image     1
## 2881:       VINTAGE BEAD COSMETIC BAG     1
## 2882:                IVY HEART WREATH     1
## 2883:                    CHAMBRE HOOK     1
## 2884:     ZINC WIRE KITCHEN ORGANISER     1
```

Visualisation

```
p <- ggplot(non_uk[, .(SALES = sum(TOTAL_SALES)), by = Country][1:10],
            aes(x = reorder(Country, SALES), y = SALES)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(title = "Top Non-UK Markets by Sales")

print(p)
```




```
ggsave("plots/non_uk_sales.png", p, width = 8, height = 6)
```

Conclusion

The UK dominates revenue and customer base, but this masks insights from international markets.

Non-UK markets such as Netherlands, EIRE, and Singapore show strong potential, with high spending behaviour and unique purchasing patterns.

Customer behaviour follows a Pareto distribution, where a small group drives most revenue.

Product demand is concentrated in decorative and gift-related items.

Business Recommendations

- Expand into high-value markets (Netherlands, EIRE, Singapore)
- Target high-value customers with loyalty programmes
- Focus on home décor and gift products
- Apply market-specific pricing strategies
- Use data-driven decision making for growth

Final Insight:

Future growth lies in international expansion and customer-focused strategies.